

Project Objective

Develop a **CRUD Management System** where an admin can manage cards and upload images/documents associated with each card.

The system should implement:

- Authentication
 - CRUD Operations
 - File Upload
 - Validation
 - Search
 - Pagination
 - Soft Delete
 - API Endpoint (Bonus)
-

Technology Stack

- PHP 8.2+
 - Laravel 12
 - MySQL
 - Bootstrap 5
 - Eloquent ORM
 - Storage File System
 - Git
-

Authentication

Use:

```
php artisan make:auth
```

or

```
php artisan breeze:install
```

Database Design

cards table

Column	Type
id	bigint
title	string
description	text
card_number	string
image	string
document	string
status	boolean
created_by	bigint
created_at	timestamp
updated_at	timestamp
deleted_at	timestamp

Migration Task

Create migration:

```
php artisan make:model Card -m
```

Migration Example

```
Schema::create('cards', function (Blueprint $table) {  
    $table->id();  
    $table->string('title');  
    $table->string('card_number')->unique();  
    $table->text('description')->nullable();  
    $table->string('image')->nullable();  
    $table->string('document')->nullable();  
    $table->boolean('status')->default(1);  
    $table->foreignId('created_by')->constrained('users');  
    $table->softDeletes();  
    $table->timestamps();  
});
```

Module 1: Authentication

Features

- Login
 - Logout
 - Dashboard
-

Module 2: Card Management

Features

Add Card

Fields:

- Title
 - Card Number
 - Description
 - Image Upload
 - Document Upload
 - Status
-

Edit Card

- Update card details
 - Replace uploaded files
-

Delete Card

- Soft Delete
-

Restore Deleted Card (Bonus)

Module 3: File Upload

Image Validation

```
'image' => 'required|image|mimes:jpg,jpeg,png|max:2048'
```

Document Validation

```
'document' => 'nullable|mimes:pdf,doc,docx|max:5120'
```

Store Files

```
$imagePath = $request->file('image')
            ->store('cards/images', 'public');

$documentPath = $request->file('document')
               ->store('cards/documents', 'public');
```

Module 4: Card Listing

Display:

- Image
 - Title
 - Card Number
 - Status
 - Created Date
 - Action Buttons
-

Module 5: Search

Search by:

- Title
 - Card Number
-

Module 6: Pagination

```
Card::latest()->paginate(10);
```

Module 7: Status Update

- Active
- Inactive

Using AJAX (Bonus).

Module 8: Validation

```
$request->validate([
    'title' => 'required|min:3|max:100',
    'card_number' => 'required|unique:cards',
    'image' => 'required|image|mimes:jpg,jpeg,png|max:2048',
    'document' => 'nullable|mimes:pdf,doc,docx|max:5120',
]);
```

Production Requirements

Create Form Request

```
php artisan make:request CardStoreRequest
```

Additional Production Features

- ✓ File Delete on Update
 - ✓ File Delete on Permanent Delete
 - ✓ UUID Card Number Generation
 - ✓ Activity Log
 - ✓ Download Uploaded Documents
 - ✓ Export CSV (Bonus)
-

Dashboard Statistics

- Total Cards
- Active Cards
- Inactive Cards
- Deleted Cards